Adapting Qwen-Image-Layered for Portrait Layer Decomposition

An Engineering and Modeling Study with CelebAMask-HQ

Zhiyuan Chen
Project Report

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Abstract

This project investigates whether `Qwen/Qwen-Image-Layered`, a model designed to decompose images into editable RGBA layers, can be adapted to a portrait-specific layer decomposition task using CelebAMask-HQ supervision. The target application is simple to state: given a portrait, produce a small number of semantically meaningful transparent layers that can be edited independently. In practice, this became a study of the gap between semantic segmentation targets and the native decomposition behavior of a pretrained layered image model. We built an end-to-end Colab pipeline for preprocessing, metadata recovery, sharded packaging, LoRA training, checkpoint comparison, and numeric evaluation. Stronger fine-tuning settings on the original five-layer target caused noisy or collapsed outputs. The later three-layer BG/HAIR/FACE formulation was more stable, even at 200 steps, but still produced only small improvements over the base model. The final 32-sample evaluation shows a small macro IoU improvement from 0.2461 to 0.2585 at checkpoint 200, driven mostly by the face layer, while layer-health diagnostics still show high overlap and frequent full-canvas layers. The main conclusion is that the engineering pipeline is reliable, but target representation and alpha separation remain the central modeling bottlenecks.

1 Motivation

Layered image editing is attractive because it turns a flat image into editable objects. If hair, face, background, text, or other components are isolated in separate RGBA layers, then downstream edits such as recoloring, moving, deleting, or replacing content can be applied locally. `Qwen-Image-Layered` is built around this idea: it takes an input image and produces multiple RGBA layers rather than a single raster output [1]. Its official examples emphasize variable-layer decomposition and editable structure.

Portraits are a natural domain for this idea. A simple portrait editor might want separate layers for background, hair, and face. CelebAMask-HQ provides pixel-level face parsing masks for 30,000 high-resolution celebrity face images, making it a reasonable source of supervision for such a task [2]. The core question for this project was therefore:

> *Can we fine-tune Qwen-Image-Layered with LoRA so that it decomposes portraits into stable, editable, semantically meaningful RGBA layers?*

The answer we found is mixed. The infrastructure can be made to work, and small numeric improvements are possible, but straightforward LoRA fine-tuning did not solve layer separation.

2 Dataset and Target Construction

2.1 CelebAMask-HQ

CelebAMask-HQ contains 30,000 face images selected from CelebA-HQ, with manually annotated 512×512 segmentation masks over facial components and accessories such as skin, hair, eyes, eyebrows, mouth, lips, glasses, earrings, neck, and clothing [2]. This annotation structure makes it useful for face parsing and facial image manipulation, but it also creates an important modeling tension: segmentation masks are crisp semantic regions, while a generative layered model may prefer broader visual chunks with soft alpha boundaries.

The raw dataset copy used in the project was not perfectly complete. Some image IDs and mask IDs did not align. The preprocessing pipeline was therefore changed to operate on the intersection of valid image IDs and valid mask IDs, recording missing files instead of failing. This turned incomplete local/Drive data into a usable training source.

2.2 Layer Export

The preprocessing pipeline exports each sample into:

- `layered_composites/`: the input portrait composite,
- `layered_layers/<sample_id>/`: target RGBA layers,
- `metadata/layered_samples.jsonl`: per-sample metadata.

For a source image $I(x, y) \in [0, 1]^3$ and a binary mask $M_k(x, y)$ for target layer k, the exported target layer is represented as an RGBA image

$$L_k(x, y) = [I(x, y) \odot M_k(x, y),\ M_k(x, y)]\,, \tag{1}$$

where the RGB channels are premultiplied by the alpha mask. This premultiplication became important later because transparent regions should not carry arbitrary hidden RGB values.

The first major target representation was a five-layer portrait scheme:

$$\mathcal{T}_5 = \{\mathrm{BG}, \mathrm{HAIR}, \mathrm{FACE_SKIN}, \mathrm{EYES}, \mathrm{MOUTH}\}. \tag{2}$$

Later, after observing weak visual improvement, we switched to a broader three-layer scheme:

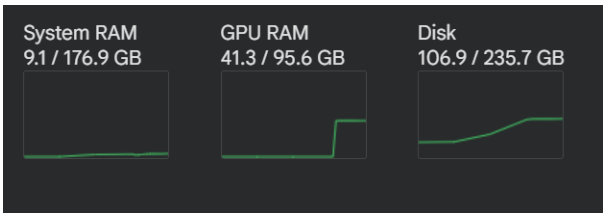
$$\mathcal{T}_3 = \{\mathrm{BG}, \mathrm{HAIR}, \mathrm{FACE}\}, \tag{3}$$

where

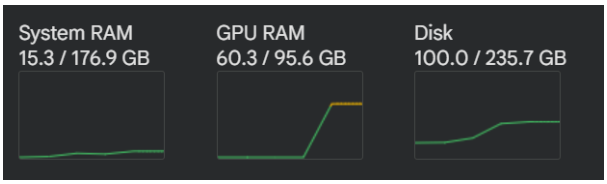
$$M_{\mathrm{FACE}} = M_{\mathrm{FACE_SKIN}} \vee M_{\mathrm{EYES}} \vee M_{\mathrm{MOUTH}}. \tag{4}$$

The background mask was defined as the complement of the foreground portrait groups:

$$M_{\mathrm{BG}} = 1 - (M_{\mathrm{HAIR}} \vee M_{\mathrm{FACE}})\,. \tag{5}$$



Training runtime



Inference/evaluation runtime

Figure 1: Colab resource snapshots for the G4-class runtime used in the GPU stages. The setup offered enough memory for the project while keeping the resource cost below the H100 runtime option available in Colab.

3 Colab and Packaging Infrastructure

The project was run with a Drive-first Colab workflow. Raw data and processed datasets lived on Google Drive, while smaller packaged artifacts were copied to local `/content` storage for GPU training. This separation mattered because Drive-backed random I/O over many small image files was too slow and fragile for training.

The main Drive layout was:

```
/content/drive/MyDrive/CelebMaskHQ_Colab/
  repo/
  raw/
  processed/
  runs/
```

Preprocessing and packaging were handled in CPU notebooks. Training, checkpoint comparison, and inference evaluation were handled in GPU notebooks. We used a paid Colab runtime exposing a G4-class GPU. The runtime reported a Blackwell-class NVIDIA RTX PRO 6000 GPU with approximately 96 GB of VRAM; this matches Google Cloud's documentation for G4 instances with NVIDIA RTX PRO 6000 Blackwell Server Edition GPUs and 96 GB GPU memory per GPU [4].

This runtime choice was also practical for Colab resource budgeting. In our Colab account, the G4 runtime used a substantially smaller compute-unit rate than the H100 option while still providing enough memory for Qwen-Image-Layered inference, LoRA training, and checkpoint comparison. The resource snapshots in Figure 1 show the training and inference/evaluation contexts used for the final experiments.

Several pieces of infrastructure were needed before meaningful modeling experiments were possible:

- metadata recovery from partially processed datasets,
- train/val sharded tar packaging,
- separate held-out test evaluation packaging,
- checkpoint save/resume logic with optimizer and scheduler state,
- LoRA adapter reactivation after resume,
- freshness checks to avoid stale Colab copies of Drive code.

This engineering work was not cosmetic. Without it, many failures looked like modeling failures but were actually stale-code, packaging, or resume issues.

Repository scope. The public repository contains the final BG/HAIR/FACE preprocessing, training, evaluation, and reporting pipeline, plus selected notebooks needed to reproduce the results discussed in this report. Earlier exploratory notebooks for the five-layer target, prompt ablations, stability reruns, and intermediate checkpoint comparisons were not included as primary artifacts because they were superseded by the final target representation and would make the repository less focused. Their conclusions are summarized in the experiment timeline and report text, while the submitted code emphasizes the reproducible final workflow.

4 Training Objective

4.1 Base Flow-Matching Loss

The training script follows the Qwen layered pipeline's latent flow-matching style objective. Let z_0 be the VAE latent of the target layered RGBA tensor and let $\epsilon \sim \mathcal{N}(0, I)$ be Gaussian noise. For a sampled noise level σ, the noisy target latent is

$$z_\sigma = (1 - \sigma) z_0 + \sigma \epsilon. \tag{6}$$

The transformer receives the packed noisy target latents together with packed conditioning latents from the input portrait. Its prediction is trained against the velocity-like target

$$v^\star = \epsilon - z_0. \tag{7}$$

Ignoring masking, the original loss has the form

$$\mathcal{L}_{\text{base}} = \mathbb{E}_{b,j} \left[w(\sigma_b) \left(\hat{v}_{\theta,bj} - v^\star_{bj} \right)^2 \right], \tag{8}$$

where b indexes batch elements, j indexes latent positions/channels/layers, and $w(\sigma)$ is the scheduler/timestep weighting from the SD3-style training utility.

4.2 Why Plain MSE Was Problematic

The target layers are sparse RGBA images. A face-feature layer or mouth layer contains only a small visible region and a large transparent canvas. In early experiments, plain latent MSE could be dominated by transparent regions or by hidden RGB values in nominally transparent pixels. This was a plausible contributor to two observed problems:

- stronger LoRA settings drifted into structured noise,
- weaker settings avoided collapse but barely changed the base model.

4.3 Alpha-Aware Modification

Two alpha-aware changes were introduced. First, target RGB values were premultiplied by alpha before training export. Second, the loss was weighted by an approximate latent-space visibility map derived from the target alpha channel.

Let $\alpha_k(x, y)$ be the target alpha mask for layer k. It is downsampled to the latent grid with max pooling:

$$\bar{\alpha}_{kj} = \mathrm{MaxPool}(\alpha_k)_j. \tag{9}$$

The latent loss weight is then

$$A_{kj} = (0.05 + 0.95\bar{\alpha}_{kj})\, V_{kj}, \tag{10}$$

where V_{kj} is a valid-layer mask. The small floor of 0.05 preserves a weak context signal for transparent regions while strongly emphasizing visible pixels.

The training loss used in the code is therefore

$$\mathcal{L}_\alpha = \frac{1}{B} \sum_{b=1}^{B} \mathrm{mean}_j \left[w(\sigma_b) A_{bj} \left(\hat{v}_{\theta,bj} - v^\star_{bj} \right)^2 \right]. \tag{11}$$

For logging, a normalized alpha-weighted MSE was also computed:

$$\mathrm{MSE}_\alpha = \frac{\sum_{b,j} A_{bj} \left(\hat{v}_{\theta,bj} - v^\star_{bj} \right)^2}{\max\left(\sum_{b,j} A_{bj}, 1 \right)}. \tag{12}$$

This modification improved stability, but it did not fully solve layer separation.

5 Experiment Timeline

5.1 Large 5-Layer Train/Val Run

The first major run used the five-layer target scheme with relatively strong LoRA settings:

- learning rate 10^{-4},
- LoRA rank 16,
- LoRA alpha 16,
- gradient accumulation 8,
- 2000 training steps,
- checkpoints every 200 steps.

The loss curve looked plausible, and validation initially suggested a useful checkpoint around step 1200. However, held-out qualitative comparison showed badly degraded outputs with colorful structured noise. This was the first major lesson: numeric training curves alone were not enough. The trained LoRA had damaged the base model's useful decomposition behavior.

5.2 Stability Rerun

The next run reduced the learning rate and step count:

- learning rate 5×10^{-5},
- LoRA rank 16,
- LoRA alpha 16,
- gradient accumulation 4,
- 800 steps,
- checkpoints every 100 steps.

The best validation checkpoint appeared much earlier, around step 200. Later checkpoints generally worsened. The outputs were less catastrophically noisy than the 2000-step run but still did not beat the base model convincingly.

5.3 Weak Micro-Ablation

To test whether the model needed only a very small update, we used:

- learning rate 10^{-5},
- LoRA rank 8,
- LoRA alpha 8,
- gradient accumulation 4,
- 100 steps,
- checkpoints every 25 steps.

This avoided collapse, but the outputs were visually close to the base model. The adaptation was stable but nearly a no-op.

5.4 Alpha-Weighted Micro-Ablation

The alpha-aware changes were then added to the weak micro-ablation recipe. This substantially improved the stability of the loss and helped avoid transparent-layer pathologies. However, the visual outcome remained modest: the LoRA changed outputs slightly but did not create reliably cleaner portrait layers.

5.5 Target Representation Change: BG/HAIR/FACE

The next hypothesis was that the five-layer target was too fine-grained and too segmentation-like. The broader target representation was:

$$\text{BG/HAIR/FACE.} \tag{13}$$

This used a 5k processed subset to reduce preprocessing overhead and keep the experiment focused on LoRA testing. The stable alpha-aware micro-ablation recipe was reused:

- learning rate 10^{-5},
- rank 8,
- LoRA alpha 8,
- gradient accumulation 4,
- 100 steps,
- validation/checkpoint every 25 steps.

The best validation result was at step 50:

checkpoint	validation loss	note
step 50	0.5089	best checkpoint
step 75	0.8571	worse
step 100	0.9682	worse
step 25	1.1364	worse

Qualitatively, checkpoint 50 gave partial improvement on some samples, especially by making one layer more hair-like or face-like, but the decomposition was still not clean. The background layer often remained nearly full-canvas, and the face layer frequently carried too much of the full portrait.

5.6 Prompt Experiments

We tested explicit layer-control prompts such as:

```
decompose this portrait into exactly three editable RGBA layers:  background,
hair, and face
```

This hurt both the base model and the LoRA in qualitative comparisons. Later inference-only prompt ablations found that the best behavior came from the default call: no explicit prompt passed, with the pipeline's default English prompt behavior enabled. Descriptive prompts such as "a close-up portrait photo of a person with visible face and hair" often reduced layer quality or collapsed slots.

5.7 Low-Learning-Rate Continuation

We also tried continuing from checkpoint 50 with a lower learning rate and linear decay. The goal was to refine the partially useful checkpoint without drifting. Instead, validation became unstable:

checkpoint	validation loss	note
step 55	0.5566	best continuation checkpoint
step 60	1.2446	worse
step 65	1.5236	worse
step 70	3.2514	severe drift
step 75	7.4991	collapse-like behavior

This reinforced the conclusion that simply training longer was not the right path.

5.8 Three-Layer 200-Step Probe

Because the earlier noisy failures came from the stronger five-layer setup, we also tested whether the newer three-layer BG/HAIR/FACE representation would remain stable at 200 steps. This run used the same 5k processed subset and the same alpha-aware settings as the 100-step BG/HAIR/FACE run, but extended the maximum step count to 200.

Unlike the old five-layer 2000-step run, the three-layer 200-step probe did not collapse into colorful noise. However, it also did not produce a decisive separation improvement. Its validation loss was still best around step 50:

validation step	validation loss	nearest saved checkpoint
50	0.5087	checkpoint 75 in the 200-step probe
75	0.8571	checkpoint 75
100	0.9680	checkpoint 100
200	0.7792	checkpoint 200
25	1.1369	checkpoint 75 in the 200-step probe

The checkpoint numbering detail matters: the 200-step probe saved checkpoints starting at step 75 in that Colab run, so the original checkpoint-50 used in the final comparison came from the earlier 100-step BG/HAIR/FACE run. The comparison therefore evaluates base model vs. original checkpoint 50 vs. new checkpoint 200.

6 Evaluation Metrics

The final numeric evaluation compared the base model, the BG/HAIR/FACE checkpoint-50 LoRA, and the BG/HAIR/FACE checkpoint-200 probe on 32 held-out test samples using the default-call inference setting.

For a predicted binary alpha mask $\hat{M}_k$ and target mask M_k, we used:

$$\mathrm{IoU}_k = \frac{|\hat{M}_k \cap M_k|}{|\hat{M}_k \cup M_k|}, \tag{14}$$

$$\mathrm{Dice}_k = \frac{2|\hat{M}_k \cap M_k|}{|\hat{M}_k| + |M_k|}. \tag{15}$$

To reduce threshold brittleness, we also used a soft alpha IoU:

$$\mathrm{SoftIoU}_k = \frac{\sum_j \min(\hat{\alpha}_{kj}, \alpha_{kj})}{\sum_j \max(\hat{\alpha}_{kj}, \alpha_{kj})}. \tag{16}$$

Alpha MAE was:

$$\mathrm{MAE}_{\alpha,k} = \frac{1}{N} \sum_j |\hat{\alpha}_{kj} - \alpha_{kj}|. \tag{17}$$

We also tracked layer-health diagnostics:

- empty-layer rate,
- full-layer rate,

- multi-layer alpha overlap,
- composite reconstruction MAE/PSNR.

The composite reconstruction metrics are only secondary. A model can reconstruct the input well by putting most of the image into every layer, so reconstruction quality does not imply editable separation.

7 Final Results

Table 1: 32-sample held-out evaluation. Higher IoU is better; lower alpha MAE is better.

model	macro IoU	soft IoU	alpha MAE	BG IoU	HAIR IoU	FACE IoU
base model	0.2461	0.2469	0.5461	0.3229	0.1607	0.2547
checkpoint-50 LoRA	0.2547	0.2554	0.5425	0.3229	0.1610	0.2800
checkpoint-200 LoRA	0.2585	0.2592	0.5512	0.3229	0.1578	0.2947

Table 2: Layer-health diagnostics. Lower is generally better for these quantities.

model	empty-layer rate	full-layer rate	multi-layer overlap
base model	0.1458	0.5208	0.8108
checkpoint-50 LoRA	0.1563	0.5313	0.8224
checkpoint-200 LoRA	0.1250	0.5313	0.8530

The LoRA improves macro IoU only modestly. Checkpoint 50 improves macro IoU by 0.0085 absolute, and checkpoint 200 improves it by 0.0123 absolute relative to the base model. The gain is concentrated in the FACE layer: FACE IoU rises from 0.2547 for the base model to 0.2800 at checkpoint 50 and 0.2947 at checkpoint 200. BG is essentially unchanged, and HAIR does not improve. Moreover, the layer-health diagnostics still show poor separation. Checkpoint 200 has the highest multi-layer overlap, 0.8530, meaning that the slight FACE IoU gain comes with worse layer separation.

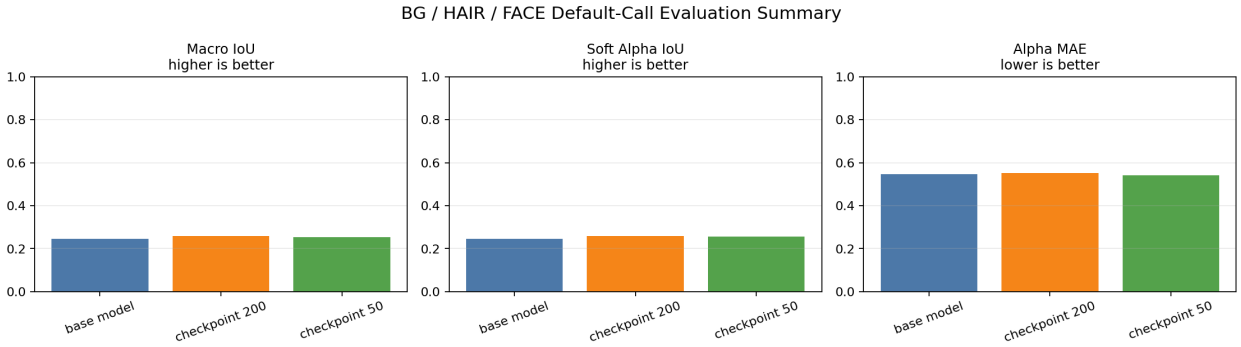


Figure 2: Aggregate metrics for base model, checkpoint-50 LoRA, and checkpoint-200 LoRA.

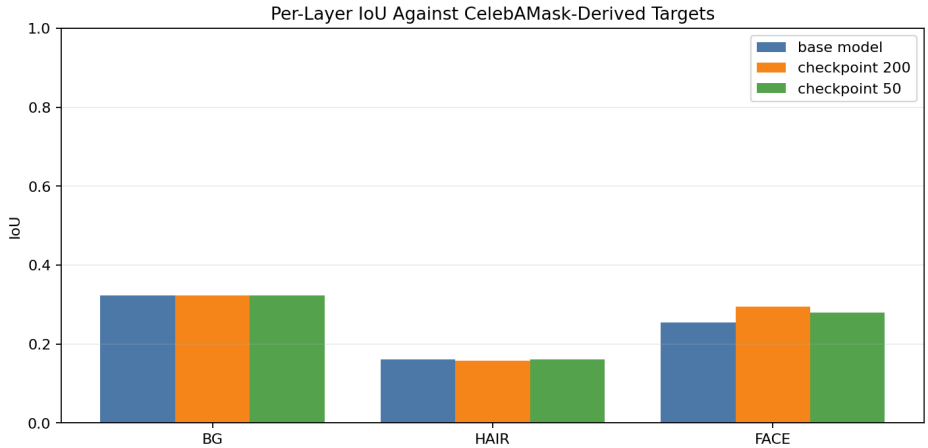


Figure 3: Per-layer IoU. The main gain is concentrated in the FACE slot; HAIR remains weak.

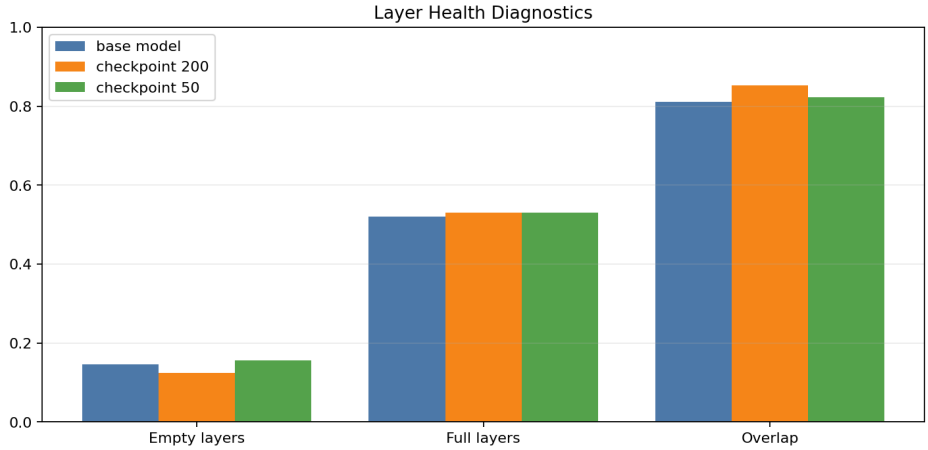


Figure 4: Layer-health diagnostics remain poor. The 200-step checkpoint does not reduce overlap; it increases the multi-layer overlap rate.

7.1 Qualitative Showcase

The qualitative comparison uses the final three-way evaluation: base model, checkpoint 50 from the earlier BG/HAIR/FACE run, and checkpoint 200 from the 200-step probe. This lets us show the updated conclusion directly: the latest three-layer setup does not collapse into obvious noise at 200 steps, but it still does not learn robust, clean portrait layer separation.

Per-sample inspection showed that the mean gain was not broad. Some samples benefited from checkpoint 200, while others degraded relative to checkpoint 50. This makes the result a weak improvement in target-mask alignment rather than a robust model-quality win.

8 Failure Cases and Lessons

8.1 More Steps Can Be Worse

The 2000-step five-layer run is the clearest noise-collapse failure case. Its loss curve did not obviously announce disaster, but visual comparison showed noisy, unusable layers. This should be understood

Figure 5: A sample where the 200-step checkpoint improves the FACE slot relative to checkpoint 50. The output is not noisy, but separation is still imperfect: the FACE layer absorbs much of the portrait and the HAIR layer remains incomplete.

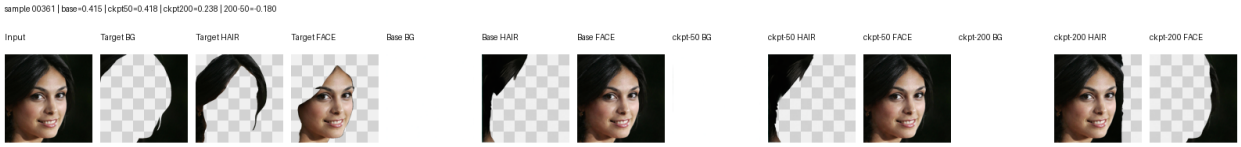


Figure 6: A sample where checkpoint 200 is worse than checkpoint 50. The longer-trained checkpoint does not become colorful noise, but it moves content into the wrong slots and reduces the macro IoU for this sample. This illustrates why 200 steps was not a reliable solution even though it remained visually stable.

as a failure of the earlier stronger five-layer setup, not as the behavior of the final three-layer setup. The newer BG/HAIR/FACE 200-step probe was visually more stable, but it still did not produce reliable separation and increased overlap. The practical lesson is therefore more specific: longer training did not monotonically improve decomposition quality, even when the target representation was simplified.

8.2 Validation Loss Did Not Fully Predict Editability

Validation loss was useful for rejecting some checkpoints, but it was not a complete proxy for visual layer quality. A checkpoint could have a better loss and still produce layers that were not meaningfully more editable than the base model. This motivated held-out visual comparison and eventually alpha-mask metrics.

8.3 Prompt Control Was Fragile

Explicit prompts that named the target layers seemed intuitive, but they worsened the decomposition. The base model behaved best when allowed to use its default prompting behavior. This suggests that Qwen-Image-Layered's prompt interface is not a simple semantic slot controller for this portrait task.

8.4 Semantic Segmentation Is Not the Same as Layer Decomposition

CelebAMask-HQ supervision is semantic and crisp. Layer decomposition for editing may be softer, broader, and more object-centric. A model trained for inherent editability may not naturally organize a face image into the same masks used by a face parsing benchmark.

9 Future Directions

The next stage should not simply train the same setup longer. More promising directions are:

Explicit alpha and separation losses. The most direct fix is to supervise alpha structure explicitly with Dice/BCE mask losses, overlap penalties, and full-canvas penalties for non-background layers. For example, an added overlap penalty could take the form

$$\mathcal{L}_{\text{overlap}} = \sum_{k<\ell} \frac{1}{N} \sum_{j} \hat{\alpha}_{kj} \hat{\alpha}_{\ell j}, \tag{18}$$

penalizing multiple layers being opaque at the same pixel.

Hybrid segmentation-guided cleanup. A practical system could combine Qwen-Image-Layered with a face parsing or segmentation model. Qwen would provide generative layer content, while a portrait segmentation prior would clean the alpha masks. This would likely improve numeric separation quickly, but it would be a hybrid system rather than pure LoRA adaptation.

More model-aligned target representations. Instead of BG/HAIR/FACE, future targets could be broader:

BG/FULL_SUBJECT/HAIR_DETAIL

or

BG/HEAD_AND_HAIR/FACE_FEATURES.

These may better match the pretrained model's native decomposition behavior.

Cleaner curated supervision. A smaller curated subset of frontal, minimally occluded portraits may reveal whether noisy or ambiguous masks are limiting learning. This would separate target noise from architectural or objective mismatch.

10 Conclusion

This project began as an attempt to fine-tune Qwen-Image-Layered into a portrait layer decomposition model. The final result is not a strong model-quality win, but it is a useful experimental finding. We built a working preprocessing, packaging, training, resume, comparison, and evaluation pipeline. We showed that aggressive LoRA training on the original five-layer target can damage the model, that weak alpha-aware LoRA training avoids that collapse, and that a broader BG/HAIR/FACE target remains stable but gives only small numeric gains. The remaining bottleneck is not basic engineering instability; it is the alignment between segmentation-derived supervision and the pretrained model's native layered representation.

References

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